

Paper 1 - Set 08 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	F	Indices and surds
2	A	Indices and surds
3	D	Quadratic functions
4	C	Quadratic functions
5	B	Simultaneous equations and inequalities
6	D	Probability
7	C	Pythagoras, mensuration and 3D
8	B	Functions and transformations
9	B	Simultaneous equations and inequalities
10	C	Pythagoras, mensuration and 3D
11	E	Polynomials, factor and remainder theorems
12	D	GCSE algebra
13	C	Circles and circle theorems
14	C	Trigonometric functions and equations
15	A	Straight lines
16	C	Exponentials and logarithms
17	B	Exponentials and logarithms
18	B	Straight lines
19	B	Circles and circle theorems
20	B	Binomial expansion

Worked solutions

Question 1 IND-01 · Indices and surds · spec MM1.1

Simplify $\frac{(2x^3)^4 \times 8x^{-5}}{4x^2}$.

Answer: F

$(2x^3)^4 = 16x^{12}$, so the numerator is $16x^{12} \times 8x^{-5} = 128x^7$. Dividing by $4x^2$ gives $32x^5$.

Question 2 IND-03 · Indices and surds · spec MM1.1

Solve $4^{x+1} = 8^{2x-3}$.

Answer: A

Write both sides as powers of 2: $2^{2x+2} = 2^{6x-9}$. So $2x+2 = 6x-9$, giving $4x = 11$ and $x = \frac{11}{4}$.

Question 3 QUA-10 · Quadratic functions · spec MM1.3

The quadratic $x^2 + px + q$ has minimum value -4 and one of its roots is 0. Which of the following could be the quadratic?

Answer: D

Minimum -4 means $x^2 + px + q = (x-h)^2 - 4$, with roots $h \pm 2$. One root is 0, so $h = \pm 2$: the quadratic is $x^2 - 4x$ or $x^2 + 4x$. Only $x^2 - 4x$ is offered.

Question 4 QUA-16 · Quadratic functions · spec MM1.3

Given that $x^2 + y^2 = 10$ and $x + y = 4$, find the value of xy .

Answer: C

$(x+y)^2 = x^2 + y^2 + 2xy$, so $16 = 10 + 2xy$ and $xy = 3$.

Question 5 SIM-17 · Simultaneous equations and inequalities · spec MM1.4

Two numbers have sum 10 and the sum of their squares is 58. What is the positive difference between them?

Answer: B

$(x-y)^2 = 2(x^2 + y^2) - (x+y)^2 = 116 - 100 = 16$, so the difference is 4 (the numbers are 3 and 7).

Question 6 PRO-09 · Probability · spec M7.4

The probabilities that a train is early, on time or late are 0.1, x and $2x$ respectively. Find the probability that the train is not late.

Answer: D

$0.1 + 3x = 1 \Rightarrow x = 0.3$; not late $= 0.1 + 0.3 = 0.4$.

Question 7 MEN-16 · Pythagoras, mensuration and 3D · spec M5.18

From the top of a cliff 40 m high, the angle of depression of a boat is 30° . How far is the boat from the foot of the cliff?

Answer: C

$$\tan 30^\circ = \frac{40}{d} \Rightarrow d = \frac{40}{1/\sqrt{3}} = 40\sqrt{3}.$$

Question 8 FUN-05 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ passes through (4, 6). Which point must lie on the graph of $y = 3f(2x) - 1$?

Answer: B

$f(2x)$ halves x -coordinates: (2, 6). Then $3f - 1$: $y = 18 - 1 = 17$. Point (2, 17).

Question 9 SIM-14 · Simultaneous equations and inequalities · spec MM1.4

Solve $2x + 3y = 12$ and $x^2 - y^2 = 0$. How many solutions (x, y) are there?

Answer: B

$x^2 = y^2$ means $y = x$ or $y = -x$. $y = x$: $5x = 12$, one solution. $y = -x$: $-x = 12$, one solution. Two in total.

Question 10 MEN-03 · Pythagoras, mensuration and 3D · spec M5.16

A trapezium has parallel sides of length 8 cm and 14 cm and area 77 cm^2 . Find the perpendicular distance between the parallel sides.

Answer: C

$$\frac{1}{2}(8 + 14)h = 11h = 77 \Rightarrow h = 7.$$

Question 11 POL-14 · Polynomials, factor and remainder theorems · spec MM1.6

$p(x)$ is a cubic polynomial with $p(1) = p(2) = p(3) = 0$ and $p(0) = 12$. Find $p(4)$.

Answer: E

$$p(x) = a(x - 1)(x - 2)(x - 3); p(0) = -6a = 12 \Rightarrow a = -2. p(4) = -2 \cdot 3 \cdot 2 \cdot 1 = -12.$$

Question 12 ALG-07 · GCSE algebra · spec M4.6

Simplify $\frac{3}{x+2} - \frac{2}{x-1}$.

Answer: D

$$\frac{3(x-1) - 2(x+2)}{(x+2)(x-1)} = \frac{x-7}{(x+2)(x-1)}.$$

Question 13 CIR-07 · Circles and circle theorems · spec MM3.3

$PQRS$ is a cyclic quadrilateral with angle $P = 3x$, angle $R = x + 40^\circ$. Find x .

Answer: C

Opposite angles of a cyclic quadrilateral sum to 180° : $4x + 40 = 180$, $x = 35^\circ$.

Question 14 TRG-01 · Trigonometric functions and equations · spec MM4.6

How many solutions does $\sin 2x = \frac{1}{2}$ have for $0 \leq x < 2\pi$?

Answer: C

$2x$ runs over $[0, 4\pi)$, which contains two full periods of $\sin$, each giving two solutions: 4 in total.

Question 15 LIN-06 · Straight lines · spec MM3.1

The perpendicular bisector of the line segment joining $(1, 5)$ and $(7, 1)$ has equation

Answer: A

Midpoint $(4, 3)$; gradient of segment $\frac{-4}{6} = -\frac{2}{3}$, so the bisector has gradient $\frac{3}{2}$:
 $y - 3 = \frac{3}{2}(x - 4) \Rightarrow 2y - 6 = 3x - 12 \Rightarrow 3x - 2y = 6$.

Question 16 LOG-05 · Exponentials and logarithms · spec MM5.2

Solve $\log_3(x + 6) + \log_3(x - 2) = 2$.

Answer: C

$(x + 6)(x - 2) = 9 \Rightarrow x^2 + 4x - 21 = 0 \Rightarrow (x + 7)(x - 3) = 0$. $x = -7$ makes both logs undefined, so $x = 3$ only.

Question 17 LOG-08 · Exponentials and logarithms · spec MM5.3

Given that $5^x = 2$ and $2^y = 125$, find the value of xy .

Answer: B

$5^{xy} = (5^x)^y = 2^y = 125 = 5^3$, so $xy = 3$.

Question 18 LIN-04 · Straight lines · spec MM3.1

The lines $y = 2x + 1$, $y = -x + 7$ and the x -axis enclose a triangle. Find its area.

Answer: B

Intersection: $2x + 1 = 7 - x \Rightarrow x = 2$, $y = 5$. x -intercepts: $-\frac{1}{2}$ and 7 , base $= \frac{15}{2}$. Area $= \frac{1}{2} \cdot \frac{15}{2} \cdot 5 = \frac{75}{4}$.

Question 19 CIR-09 · Circles and circle theorems · spec MM3.2

Find the length of the tangent from the point $(7, 1)$ to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$.

Answer: B

Centre $(1, 2)$, radius 3 . Distance from $(7, 1)$ to centre: $\sqrt{36 + 1} = \sqrt{37}$. Tangent length $= \sqrt{37 - 9} = \sqrt{28}$.

Question 20 BIN-14 · Binomial expansion · spec MM2.4

How many terms in the expansion of $(x + y + z)^4$ (after collecting like terms) are there?

Answer: B

Terms are $x^a y^b z^c$ with $a + b + c = 4$, $a, b, c \geq 0$. Counting: for $a = 0$ there are 5 choices of (b, c) , $a = 1$: 4, ..., $a = 4$: 1. Total $5 + 4 + 3 + 2 + 1 = 15$.